

The Bimodal Manifold Interaction (BMI) Framework

A Generative Geometric Model of Particle Resonance and Cosmological Coalescence

A Case Study in Human-AI Collaborative Theoretical Physics and Meta-Analysis

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Environment: GitHub BMI Research Lab / Local Copy Bridge

Abstract

The origin of fundamental particle masses, spin, and the macroscopic expansion of the universe remain primary unresolved fine-tuning problems in the Standard Model, traditionally requiring arbitrary empirical constants. This paper introduces the Bimodal Manifold Interaction (BMI) framework, a purely geometric, topological model that reformulates both quantum particle physics and macroscopic cosmology as functions of manifold interaction and resonance.

At the microscopic scale, by modeling particles as phase-locked geometric configurations of nested tori, we demonstrate that the masses of the electron, neutrino, muon, pion, and nucleons emerge deterministically as integer harmonic modes of a unified base frequency spectrum. Properties such as spin and charge are shown to be emergent topological invariants arising from phase-alignment rather than ad hoc inputs.

At the macroscopic scale, the BMI framework redefines cosmic expansion not as dark energy acting on matter, but as the coalescence of two fundamental branes (σ_A and σ_B). By modeling the system through dynamic **Bridge Stress** (τ) and **Inter-brane Potential** (V_{gap}), the simulation successfully retrodicts the high-energy epoch of the Cosmic Microwave Background (CMB) and mathematically predicts a cosmic destiny of **Asymptotic Flatness**—the final geometric alignment of the interacting manifolds.

Parallel to the theoretical physics framework, this project serves as a rigorous meta-analysis evaluating the validity, boundaries, and methodology of utilizing Large Language Models (LLMs) as advanced research assistants. The successful derivation of the BMI spectrum and the resolution of human-AI "state drift" within a version-controlled pipeline demonstrates that structured human-AI collaboration can independently formulate, audit, and systematically advance paradigm-shifting scientific models.

Core Harmonic Mass Representation

To bridge the gap between deterministic scripts and manuscript compilation across systems, the foundational microscopic resonance governing the spectrum is structured as a parameterized function. The raw mass relation maps to the localized manifold energy according to the following baseline expression:

$$M_n = n \times \omega_0 \cdot (1 + \Delta\psi_{\text{harmonic}})$$

Code Realization Bridge:

- **Python Expression:** `M_n = n * omega_0 * (1 + delta_psi)`
- **LaTeX Source:** `M_n = n \times \omega_0 \cdot (1 + \Delta\psi_{\text{harmonic}})`

Where:

- M_n represents the derived rest mass of the targeted harmonic mode.
- n belongs to the set of positive integers representing the structural mode index.
- ω_0 constitutes the fundamental base frequency of the bimodal manifold.
- $\Delta\psi_{\text{harmonic}}$ dictates the local geometric phase-alignment deviation.

Project Goals & Meta-Analysis Framework

The BMI Project operates on two distinct, synergistic tracks: a **Primary Physics Track** defining a unified cosmological/quantum model, and a **Secondary Meta-Analysis Track** auditing AI-assisted scientific discovery.

1. Primary Physics Track: The Geometric Lab

The analytical objective is to transition physics from empirical cataloging to a generative, topological architecture. Current focus areas within the GitHub research lab include:

- **Spectral Mapping:** Cataloging the linear resonance spectrum of known particles and probing the harmonic "mass gaps" to isolate unobserved topological modes.
- **Cosmological Evolution:** Utilizing the BMI Engine to simulate manifold coalescence, validating temporal invariance via dynamic thermal amplitude scaling (The Cradle Test) and asymptotic zero-state equilibrium.
- **Impedance and Forces:** Modeling forces not as independent fields, but as manifold impedance and overlap integrals when resonant geometries interact.

2. Secondary Track: The AI-Collaborative Meta-Analysis

This project acts as an explicit stress-test of generative AI's capacity to handle professional-grade theoretical development. The methodology log documents:

- **Overcoming State Drift:** Transitioning from iterative, per-line code edits to an "Atomic File Overwrite" methodology to eliminate interpretative divergence between the human researcher and the AI.
- **The Validation Pipeline:** Utilizing a multi-stage verification framework where structural, dimensional, and logical consistencies are mathematically verified before integration.
- **Audit Trail Integrity:** Maintaining a version-controlled GitHub repository to ensure every theoretical leap (e.g., the transition from Observational Particle Cosmology to Topological Manifold Coalescence) is transparently documented and reproducible.